Circuits, Systems, and Signal Processing (2020) 39:2673–2694
https://doi.org/10.1007/s00034-019-01285-w

Optimal Nonlinear Signal Approximations Based on Piecewise Constant Functions

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Received: 3 June 2018 / Revised: 8 October 2019 / Accepted: 10 October 2019 /
Published online: 14 October 2019
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Abstract

We provide here an optimal method of approximating a signal by piecewise constant functions. To this end, we minimize over the signal subdomains a fidelity term between the signal and its corresponding piecewise approximations; subdomains being determined by the number of approximations samples used for. An optimal recursive relationship is then obtained and proven, which helps us to derive the proposed approximation algorithm. The complexity of the algorithm is $O(MN^2)$, where N is the number of samples of the processed signal and M is the number of piecewise constant approximation functions. There are different techniques to approximate a signal using piecewise constant functions, wavelet decomposition is one of them by means of a Haar wavelet. Our approach is then compared to linear and nonlinear wavelet-based approximations, and both qualitative and quantitative results are provided on various tested signals, showing the efficiency of the proposed approach.

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Keywords Nonlinearity · Optimization · Approximations · Piecewise constant functions · Wavelets

1 Introduction

Most real-world problems can not be solved explicitly, and approximate solutions are preferred to exact solutions that are computationally not feasible. Thus, approximation techniques have been for a while a subject of interest, due to demanding needs ranging from practical to theoretical concerns. Approximation methods have been applied in many domains: image compression, audio coding, learning theory, filters/transforms, and compressed sensing [10,11,17,19,28,39–41,43,56] to name a few. Approximation theory aims at resolving the problem of a possibly complicated function, usually called the target function, by means of simple and easy-to-compute functions often called the approximants. Therefore, the fundamental question regarding the trade-off between the resolution and the complexity of the approximation [14] naturally raises. Thus, mathematical and numerical analyses are needed for a rigorous framework of approximation algorithms and can lead to useful advances in the field. A brief review of both the theory of nonlinear approximation techniques and their applications can be found in [15]. Spline functions [5,13,38,63,64] has been one of the most famous approximation tools that have been studied in the recent years, mainly because of their suitability and convenience, and also they provide optimal theoretical estimation solutions from limited data. A good review of seminal approximations theory and some of the related methods can be found in [53], wherein standard polynomial functions, best approximations from linear space with respect to the minimax, least squares and L^1 norms were described and analyzed.

In mathematical approximation theory, an accurate signal approximation basis is constructed with a linear combination of a small number of vectors selected within that basis. Orthogonal bases are, therefore, very important in signal processing and analysis, since they can efficiently be used to approximate signals with just a few number of vectors. Many techniques have been developed using classical Fourier analysis [4] or wavelets [44,46,52]; however, there are other approaches studied over the years as well. For example, approximations were performed using superpositions of sigmoidal functions [12]; in a related study, an approximation approach based on data-adaptive normalized Gaussian functions was proposed [54], while adaptive chirplet was used in [55] and a gradient descent 'boosting' paradigm [24]. An optimal approximation with piecewise smooth functions was proposed by Mumford and Shah [49] in two dimensions in a variational setting. More recently, sparse approximate solutions were considered by [45] which can be efficiently solved with various techniques such as the least absolute shrinkage and selection operator [62], compressed sensing [19], basis pursuit and separability [69], learning sparse dictionaries [57], and more recently using Bregman iterations [31], and non-separable regularization [59]. Weng and Barner [68] proposed an optimal reconstruction based on a data-driven method called the empirical mode decomposition (EMD) [30]. An interesting approach was proposed by Serra et al. [60] recently based on a fast variational inference approach in order to build a

well-suited overcomplete dictionary with redundant set of atoms. They also found a sparse representation of a given signal with respect to a chosen dictionary.

1.1 Related Works

In this work, we are interested in piecewise constant approximation, which is particularly interesting due to the various applications in signal and image processing areas. We review here some of the related signal approximation algorithms and applications. A recursive Bayesian estimation approach was proposed in [37], and this approach was compared to the Bayesian inference approach in piecewise constant Gaussians [47] and stochastic models [65]. Among other approaches that were proposed for piecewise constant approximations, we mention [35] which proposed signals recovery using compressed sensing , l_0 penalized estimation on graphs [22], and group-fused graphical Lasso [25]. For multidimensional signals, an approach using proper orthogonal decompositions was proposed in [66] and using total variation by [23]. Interesting applications of such approximations can be found, e.g., in [70], where a piecewise constant and low-rank approximation is used in identifying recurrent copy number variations in genetic data and using an approximation method to determine the amplitude levels of piecewise constant signals [67].

In the context of interesting applications, a sparsity and dictionary learning image denoising method was proposed by [20] and a feature reduction method for hyperspectral images by [33]. Mean-field annealing minimization was proposed in [29] for image restoration/reconstruction, a reconstruction based on nonsmooth and nonconvex minimization approach by [50], and on nonuniform Fourier samples using a convex matrix completion algorithm [51]. A Bayesian algorithm was developed by [18] for joint segmentation of piecewise constant autoregressive processes, and similarly, a segmentation technique based on dual constraint relaxation was proposed by [48]. In a recent work of Shah et al. [61], a hyperbolic tangent-based surrogate function was proposed for image segmentation to approximate the l_1-norm regularization. A dynamic programming (DP) approach was proposed in a classical work of Bellman and Roth [3] for best fitting a curve with a sequence of straight line segments, by minimizing the maximum absolute deviation between the fitted curve and the processed data. In another classic work by Konno and Kuno [36], a best piecewise constant approximation of a one-variable function was presented, by finding the best piecewise constant function among all possible piecewise constant functions that are defined in the same domain according to the Euclidean and sup norms, on one hand, and meanwhile minimizing the number of steps, on another hand. Any piecewise continuous function was then approximated by a piecewise constant function using an average between the upper and lower envelopes of the corresponding processed function.

Previously proposed algorithms were based primarily on the shortest path and DP methods. An approach based on best fitting methods was proposed by Auger et al. [2]. A DP technique was used for finding a finite number of non-overlapping segment neighborhoods which were identified by optimally partitioning a sequence into contiguous segments; a minimization of the sum of the fit measures was then applied to carry out the final step of the approximation. A recursive algorithm was proposed by

Jackson et al. [32] for finding an optimal partition of the data on an interval similar to the landmark work of DeVore [14]; the fitness function should fulfill, however, an additive property requirement; a DP method was proposed for its implementation. The closest work of our proposed method here is the one presented in [14]. However, the classical methods presented there were addressed to continuous functions, formulated in a continuous domain, and discretized to obtain proper numerical estimates. Further, the problem was both formulated for linear, nonlinear, and adaptive approximations. The target function was defined on an interval and approximated in different manners by piecewise constant with finite pieces. It was shown that functions having some approximation orders could thus be characterized and such characterizations enlightened, for instance, the distinctions between linear and nonlinear approximations.

1.2 Main Contributions

It is well known that nonlinear approximations based on wavelet decompositions are better than the corresponding linear ones. Also, in nonlinear approximations, approximants truly come from nonlinear manifolds, instead of linear spaces. Advantages of nonlinear approximations over the simpler and more established linear methods were explored in [14], by studying the approximation rate which was found to be decreasing in error regarding the number of parameters in the approximant. Our main contributions here can be summarized as follows:

1. Direct formulation of the problem in a discrete domain, contrary to the continuous setting considered in [14].
2. Proposition of a nonlinear optimal approach to approximate a signal with piecewise constant functions: this is done by minimizing over the signal subdomains a fidelity term expressed as the minimum squared error (MSE) between the signal and its corresponding piecewise approximations; subdomains are, in fact, determined by the number of approximations samples used for.
3. Establishment and proof of an optimal recursive relationship.
4. Derivation of an approximation algorithm: We denote some similarities in the formulation with the quantization method considered in [7]. The complexity of the proposed algorithm is of order $O(MN^2)$, where N is the length of the processed signal and M is the number of piecewise constant approximation functions, $M<N$. The number of parameters is correlated to the computational effort, as noted in [14]. Piecewise constant approximations can also be obtained using Haar-based wavelet decomposition, but not in an optimal way as proposed in our approach.
5. Good qualitative and quantitative performances with our approach are in comparison with linear and nonlinear approximations based on Haar wavelet. Haar functions are also important in averaging theory [58], roughly meaning approximating solutions of differential systems by a solution of an *average* system. A relationship was established in [9] with local averages given by a Haar approximation, and the averaging relation was recovered when the product was defined by a two-point Hermite interpolation. Also, note that the Haar wavelet is simple and amenable to digital signal processors.

1.3 Organization

We organize the rest of the paper as follows. Section 2 presents some basics of wavelets and approximations. Our approach is presented in Sect. 3, while the algorithm and numerical results are proposed in Sect. 4. Conclusion and some perspectives of this work are provided in Sect. 5.

2 Background on Approximations and Wavelets

2.1 Continuous Piecewise Constant Approximations

Linear approximation Let $N \in \mathbb{N}^\star$ and T be an ordered set of points $0 = t_0 < t_1 < \cdots < t_N = 1$ in $\Omega = [0,\ 1]$. Such points form a partition $\Pi = \{I_k\}_{k=1}^N$ of Ω into N disjoint intervals $I_k = [t_{k-1};\ t_k]$, $1 \le k \le N$. Denote $\mathscr{S}_1(T)$ the space of piecewise constant functions relative to this partition, the family of characteristic functions $\{\chi_I : I \in \Pi\}$ is a basis for $\mathscr{S}_1(T)$, meaning that $\mathscr{S}_1(T)$ is a linear space of dimension N. For $0 < p \le \infty$, the following approximation error of $f \in L^p(\Omega)$ by elements of $\mathscr{S}_1(T)$ was introduced [14]:

$$s(f, T)_p = \inf_{S \in \mathscr{S}_1(T)} \|f - S\|_{L^p(\Omega)}. \tag{1}$$

Properties of f and T that determine $s(f, T)_p$ were then studied.

For $0 < \alpha \le 1$, $0 < p \le \infty$, and $M > 0$, denote $Lip(\alpha, L^p(\Omega))$ the set of functions $f \in L^p(\Omega)$ *s.t.* $\|f(\cdot + h) - f\|_{L^p([0;\ 1-h])} \le Mh^\alpha$, $0 < h < 1$, endowed with the norm $|f|_{Lip(\alpha, L^p(\Omega))} = \max_{t>0} t^{-\alpha} \left\{ \max_{|h| \le t} \| \sum_{k=0}^{r} (-1)^{r-k} \binom{r}{k} f(\cdot + kh) \|_{L^p} \right\}$. Let us define $S_T \in \mathscr{S}^1(T)$ *s.t.*:

$$S_T(x) = \frac{1}{|I|} \int_I f \mathrm{d}t,\ x \in I,\ I \in \Pi, \tag{2}$$

where $|I|$ denotes the Lebesgue measure of I. Then, one has [16]

$$s(f, T)_p \le \|f - S_T\|_{L^p(\Omega)} \le C_p |f|_{Lip(\alpha, L^p(\Omega))} \delta_T^\alpha, \tag{3}$$

where C_p is a constant that depends at most on p and $\delta_T = \max_{0 \le k < N} |t_{k+1} - t_k|$.

Let $\triangle_n = \{\frac{k}{n};\ 0 \le k \le n\}$ be a set of n equally spaced points from Ω, we have the following result:

Proposition 1 [14] *If f satisfies $s(f, \triangle_n)_p \le Mn^{-\alpha}$, $n = 1, 2, \cdots$, for some $0 < \alpha \le 1$ and $M > 0$, then $f \in Lip(\alpha, L^p(\Omega))$ and $|f|_{Lip(\alpha, L^p(\Omega))} \le C_p M$.*

Nonlinear approximation Let T be a finite set of points $0 = t_0 < t_1 < \cdots < t_n = 1$ from $\Omega = [0,\ 1]$, and denoted by $\mathscr{S}^1(T)$ the set of functions S which are piecewise constant with breakpoints from T. Let $\sum_n$ be the nonlinear space defined as $\sum_n =$

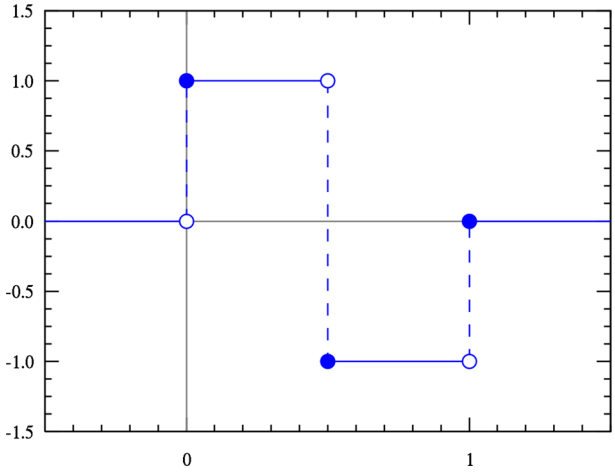


Fig. 1 Haar wavelet

$\bigcup_{\#T=n+1} \mathscr{S}^1(T)$, with $\#T$ representing the cardinality of T. Then, each function in $\sum_n$ is piecewise constant with at most n pieces. For any $f \in L^p(\Omega)$, $0 < p \leq \infty$, the L^p nonlinear approximation error $\sigma_n(f)_p$ of f is introduced:

$$\sigma_n(f)_p = \inf_{S \in \sum_n} \|f - S\|_{L^p(\Omega)}. \tag{4}$$

where $\sum_n$ is truly a nonlinear space. The following result [34] is proven in [14]:

Proposition 2 *For any $f \in C(\Omega)$, one has*

$$\sigma_n(f)_\infty \leq \frac{M}{2n}, \ n = 1, 2, \ldots, \tag{5}$$

if and only if $f \in BV(\Omega)$[1].

2.2 Basics on Wavelets

Definition 1 – A wavelet is a function $\psi \in L^2(\mathbb{R})$ with a zero mean, *i.e.*,

$$\int_{\mathbb{R}} \psi(t)\mathrm{d}t = 0, \tag{6}$$

[1] For $f \in L^1(\Omega)$, denote $\int_\Omega |Df| = \sup\left\{\int_\Omega f \operatorname{div}(\varphi)\mathrm{d}x;\ \varphi = (\varphi_1, \varphi_2) \in C_0^1(\Omega, \mathbb{R}^2) \text{ and } \|\varphi\| \leq 1\right\}$, where $C_0^1(\Omega)$ is the space of continuously differential functions with compact support in Ω, $\operatorname{div}(\varphi) = \frac{\partial \varphi_1}{\partial x_1} + \frac{\partial \varphi_2}{\partial x_2}$ and derivatives are taken in a distributional sense, $\|\varphi\| = \|(\varphi_1^2 + \varphi_2^2)^{1/2}\|_{L^\infty(\Omega)}$. $\mathrm{BV}(\Omega)$ is the set of functions defined by $BV(\Omega) = \left\{f \in L^1(\Omega);\ s.t.\ \int_\Omega |Df| < \infty\right\}$. See [1,21,26] for more details on BV spaces.

normalized, *i.e.*, $\|\psi\|_{L^2} = 1$, and centered in the neighborhood $\mathscr{V}_{t=0}$ of $t = 0$.
– The wavelet transform $Wf(u, s)$ of a function $f \in L^2(\mathbb{R})$ at time t and scale u is given by

$$Wf(u, s) = \int_{\mathbb{R}} f(t) \frac{1}{\sqrt{t}} \psi^{\star}\left(\frac{t-u}{s}\right) \mathrm{d}t. \tag{7}$$

The following theorem [8,27] guarantees a real wavelet transform completeness and an energy conservation property:

Theorem 1 *Let $\psi \in L^2(\mathbb{R})$ be a real function, i.e.,*

$$C_{\psi} = \int_{\mathbb{R}^+} \frac{|\hat{\psi}(\omega)|^2}{\omega} \mathrm{d}\omega < +\infty, \tag{8}$$

where $\hat{\psi}$ is the Fourier transform of ψ. Then, any function $f \in L^2(\mathbb{R})$ satisfies

$$f(t) = \frac{1}{C_{\psi}} \int_{\mathbb{R}^+ \times \mathbb{R}} Wf(u, s) \frac{1}{\sqrt{s}} \psi\left(\frac{t-u}{s}\right) \mathrm{d}u \frac{\mathrm{d}s}{s^2} \text{ and} \tag{9}$$

$$\int_{\mathbb{R}} |f(t)|^2 \mathrm{d}t = \frac{1}{C_{\psi}} \int_{\mathbb{R}^+ \times \mathbb{R}} |Wf(u, s)|^2 \mathrm{d}u \frac{\mathrm{d}s}{s^2}. \tag{10}$$

Example 1 The piecewise constant and discontinuous Haar wavelet is given by

$$\psi(t) = \begin{cases} 1, \text{ if } t \in [0;\ \frac{1}{2}[\\ -1, \text{ if } t \in [\frac{1}{2};\ 1[\\ 0, \text{ elsewhere} \end{cases}, \tag{11}$$

and is represented in Fig. 1.

A wavelet ψ can be constructed in a way such that the dilated and translated family $\{\psi_{j,n}(t)\}_{(j,n)\in\mathbb{Z}^2}$, given $\forall\, n \in \mathbb{Z}$ and $\forall\, j \in \mathbb{Z}$ by

$$\psi_{j,n}(t) = \frac{1}{\sqrt{2^j}\,\psi\left(\dfrac{t - 2^j n}{2^j}\right)},$$

is an orthonormal basis of $L^2(\mathbb{R})$ [44].

2.3 Approximations in Hilbert Spaces

Let us consider an orthonormal basis $\mathscr{B} = \{\beta_n\}_{n\in\mathbb{N}}$ of a Hilbert space H. Then, for any function $f \in H$, one has $f = \sum_{n\in\mathbb{N}} < f, \beta_n > \beta_n$, where $<\cdot, \cdot>$ is the scalar product in H.

Linear approximation A linear approximation f_K of f consists of only considering the first K terms, *i.e.*,

$$f_K = \sum_{n=0}^{K-1} < f, \beta_n > \beta_n, \tag{12}$$

which can be viewed as an orthogonal projection of f on the space generated by $\{\beta_n\}_{n=0}^{K-1}$. The linear approximation error is then

$$\|f - f_K\|^2 = \sum_{n \geq K} | < f, \beta_n > |^2. \tag{13}$$

For a wavelet basis adapted to $L^2([0;\ 1])$ with compactly supported C^q wavelets with q vanishing moments, *i.e.*,

$$\mathscr{B} = \left[\{\phi_{J,n}\}_{0 \leq n \leq 2^{-J}}, \{\psi_{j,n}\}_{-\infty < j \leq J}^{0 \leq n \leq 2^{-j}} \right], \tag{14}$$

f_K is then given by [44]

$$f_K = \sum_{n=0}^{2^{-l}-1} < f, \phi_{l,n} > \phi_{l,n}, \tag{15}$$

with $K = 2^{-l}$. The linear approximation error is thus [44]

$$\|f - f_K\|^2 = \sum_{j \leq l, n=0}^{2^{-j}-1} | < f, \psi_{j,n} > |^2. \tag{16}$$

Theorem 2 *Let $0 < s < q$ be a Sobolev exponent. If $f \in L^2([0;\ 1])$ is in the Sobolev space $W^s[0;\ 1]$*[2]*, then the linear approximation error is $o(K^{-2s})$.*

Nonlinear approximation Let us consider K vectors selected in $\mathscr{B}$ and denote I_K the set of indices of such vectors. A nonlinear approximation f_K of f is then the projection

$$f_K = \sum_{n \in I_K} < f, \beta_n > \beta_n, \tag{17}$$

and the nonlinear approximation error is then

$$\|f - f_K\|^2 = \sum_{n \notin I_K} | < f, \beta_n > |^2. \tag{18}$$

[2] $W^s[0;\ 1]$ is the space of functions which are s times differentiable is the sense of Sobolev over the domain $[0;\ 1]$. A good review of Sobolev spaces can be found in [6].

By rewriting basis elements (14) $\phi_{J,n} = \psi_{J+1,n}$, the best nonlinear approximation f_K of $f \in L^2([0;\ 1])$ from K wavelets is given [44] by

$$f_K = \sum_{(j,n)\in I_K} < f, \psi_{j,n} > \psi_{j,n}. \tag{19}$$

Of course, I_K represents here the set of indices of the K wavelet coefficients with the largest amplitude $| < f, \psi_{j,n} > |$. The nonlinear approximation error is thus

$$\|f - f_K\|^2 = \sum_{(j,n)\notin I_K} | < f, \psi_{j,n} > |^2. \tag{20}$$

The nonlinearity comes from the set I_K; therefore, Eq. (20) is totally different from (16) and explains also the improvements of the nonlinear approximation over the linear approximation.

Proposition 3 [44] *Assuming that f has a finite number of discontinuities in* $[0;\ 1]$ *and is uniformly Lipschitz $\alpha < q$ between discontinuities. Then, the nonlinear approximation error* (20) *is $O(K^{-2\alpha})$.*

The preceding theoretical results show that the regularity of the signal is important in order to infer the involved approximation errors. Rather than going from the continuous to the discrete case, we start by discretizing the continuous signal to a finite samples signal; then, we derive an algorithm. This is another contribution of the proposed discrete approach for which the only restriction is to constraint the processed signal to have a finite number of samples, rather than going from the continuous. The theoretical results also show that the proposed approach is optimal; however, its computation time is greater than the traditional wavelets-based approximations. This is explained by the fact that the number of parameters is correlated to the computational effort, as noted previously in [14].

3 Optimal Piecewise Constant Nonlinear Approximation

We also denote $[1;\ n]$ the set $\{1, 2, \cdots, n\}$, for all $n \in \mathbb{N}^\star$. For any discrete signal f, $f[n]$ represents its samples; for any continuous signal f, its values are denoted by $f(t)$ where $[1,\ n]$ is an interval of $\mathbb{R}$ as usual.

Let $S : [0;\ N] \longrightarrow \mathbb{R}; n \mapsto S[n]$, be a discrete signal of size N. $\forall\ p \in [0;\ N]$ and $\forall\ k \in [0;\ N]$, let us denote by $S_{p,k}[n]$ the piecewise constant signal in $[0;\ k]$ taking at most p different values and minimizing the following l^2 square norm:

$$\varepsilon_{p,k} = \|S - S_{p,k}\|^2_{[0;k]} = \sum_{n=0}^{k} |S[n] - S_{p,k}[n]|^2. \tag{21}$$

$\varepsilon_{p,k}$ is, in fact, the approximation error of S in $[0;\ k]$, since it measures the difference between S and the piecewise constant function $S_{p,k}$.

Proposition 4 *$\forall\, k \in [0;\ N]$ and $\forall\, l \in [0;\ k-1]$, let us define values $a_{l,k}$, and let us consider $c_{l,k}$ defined as*

$$c_{l,k} = \sum_{n=l}^{k} |S[n] - a_{p,k}|^2. \tag{22}$$

Then, constant values $a_{l,k}$ minimizing $c_{l,k}$ are given by

$$a_{l,k} = \frac{1}{1+k-l} \sum_{n=l}^{k} S[n]. \tag{23}$$

Proof The proof is straightforward by considering the differentiable mapping:

$$\begin{aligned} c :& \mathbb{R} \longrightarrow \mathbb{R}_+ \\ & x \mapsto c(x) = \sum_{n=l}^{k} |S[n] - x|^2. \end{aligned} \tag{24}$$

Thus, $a_{l,k}$ is a minimum of c if and only if $c'(a_{l,k}) = 0$, which results in

$$a_{l,k} = \frac{1}{1+k-l} \sum_{n=l}^{k} S[n]. \tag{25}$$

□

The following result gives a recurrence formula on how to calculate approximation errors $\varepsilon_{p,k}$, which make the proposed piecewise constant method optimal:

Proposition 5

$$\varepsilon_{p,k} = \min_{l \in [0;\ k-1]} [\varepsilon_{p-1,l} + c_{l+1,k}]. \tag{26}$$

Proof $\forall\, l \in [0;\ k-1]$, one has

$$\varepsilon_{p-1,l} + c_{l+1,k} = \sum_{n=0}^{l} |S[n] - S_{p-1,l}[n]|^2 + \sum_{n=l+1}^{k} |S[n] - a_{l+1,k}|^2. \tag{27}$$

Let us consider the piecewise constant mapping $g : [0;\ k] \longrightarrow \mathbb{R}$ defined by

$$g[n] = \begin{cases} S_{p-1,l}, \text{ if } 0 \le n \le l \\ a_{l+1,k}, \text{ if } l+1 \le n \le k \end{cases} \tag{28}$$

g takes at most p values on $[0;\ k]$, since $S_{p-1,l}$ takes at most $p-1$ values, and $a_{l+1,k}$ is a constant. Thus, by construction, one has

$$\sum_{n=0}^{l} |S[n] - S_{p-1,l}[n]|^2 + \sum_{n=l+1}^{k} |S[n] - a_{l+1,k}|^2$$
$$= \sum_{n=0}^{l} |S[n] - g[n]|^2 + \sum_{n=l+1}^{k} |S[n] - g[n]|^2, \tag{29}$$

if and only if $\varepsilon_{p-1,l} + c_{l+1,k} = \sum_{n=0}^{k} |S[n] - g[n]|^2 = \|S - g\|^2_{[0;\ k]}$. Because $f_{p,k}$ minimizes that norm, then it follows

$$\|S - g\|^2_{[0;\ k]} \geq \|S - S_{p,k}\|^2_{[0;\ k]} = \varepsilon_{p,k}. \tag{30}$$

Thus, one has

$$\forall\, l \in [0;\ k-1],\ \varepsilon_{p,k} \leq \varepsilon_{p-1,k} + c_{l+1,k}. \tag{31}$$

Since we operate on a discrete and finite set, then $\varepsilon_{p,k}$ is the maximal element of the finite family $\{\varepsilon_{p-1,k} + c_{l+1,k}\}_{l=0}^{k-1}$. Thus, one has

$$\varepsilon_{p,k} = \min_{l \in [0;\ k-1]} [\varepsilon_{p-1,l} + c_{l+1,k}]. \tag{32}$$

□

4 Algorithm and Results

4.1 Piecewise Constant Approximation Algorithm

Using relation (21), Eq. (26) in Proposition 5 is equivalent to

$$\sum_{n=0}^{k} |S[n] - S_{p,k}[n]|^2 = \min_{l \in [0;\ k-1]} \left[\sum_{n=0}^{l} |S[n] - S_{p-1,l}[n]|^2 + \sum_{n=l+1}^{k} |S[n] - a_{l+1,k}|^2 \right]. \tag{33}$$

Let $M \in \mathbb{N}^\star$, $M \leq N$, and fixed. We propose below a way to calculate $S_{M,N}$ by iteratively computing $S_{p,k}$, $\forall\, p \in [1;\ M]$, and $\forall\, k \in [0;\ N]$.
Due to our definitions of $S_{p,k}$, $S_{0,k}$ does not exist since it takes 0 value in $[0;\ k]$. Also, $S_{p,0}$ is only defined at 0, and thanks to $\varepsilon_{p,0}$, one has

$$\forall\, p \in [1;\ M],\ S_{p,0} = S(0). \tag{34}$$

$\forall\ k \in [0;\ N]$, $S_{1,k}$ takes at most one constant value in $[0;\ k]$ given by (23) in Proposition 4, *i.e.*,:

$$\forall\, k \in [0;\ N],\ S_{1,k} = \frac{1}{k+1}\sum_{n=0}^{k} S[n] := a_{0,k}. \tag{35}$$

Before generalizing, let us show the case for $p = 2$. We then need to determine the value denoted $l_2(k)$ (since l depends on both p and k) for which the sum $\varepsilon_{1,l} + c_{l+1,k}$ over all $l \in [0;\ k-1]$ is minimum:

$$\sum_{n=0}^{k} |S[n] - S_{2,k}[n]|^2 = \sum_{n=0}^{l_2(k)} |S[n] - S_{2,k}[n]|^2 + \sum_{n=l_2(k)+1}^{k} |S[n] - S_{2,k}[n]|^2 \tag{36}$$

$$= \sum_{n=0}^{l_2(k)} |S[n] - a_{0,l_2(k)}|^2 + \sum_{n=l_2(k)+1}^{k} |S[n] - a_{l_2(k)+1,k}|^2. \tag{37}$$

By identification, one has

$$S_{2,k}[n] = \begin{cases} a_{0,l_2(k)}, \text{ if } 0 \le n \le l_2(k) \\ a_{l_2(k)+1,k}, \text{ if } l_2(k)+1 \le n \le k \end{cases}, \tag{38}$$

where $a_{l_2(k)+1,k}$ are given by (23) in Proposition 4.

The case $p = 3$ is obtained in the same way as above, as follows:

$$S_{3,k}[n] = \begin{cases} S_{2,l_3(k)}[n], \text{ if } 0 \le n \le l_3(k) \\ a_{l_3(k)+1,k}, \text{ if } l_3(k)+1 \le n \le k \end{cases}, \tag{39}$$

where $S_{3,k}[n]$ is given by (38) with $k = l_3(k)$ and $a_{l_3(k)+1,k}$ are given by (23) in Proposition 4. It is easy to notice that $S_{3,k}[n]$ takes three values on $[0;\ k]$. Before generalizing, let us show this example to illustrate how our approximation method works:

Example 2 Let us consider the following continuous signal:

$$S : [0;\ 30] \mapsto \mathbb{R};\ x \longrightarrow S(x) = \begin{cases} 1 + \cos(2\pi x), \text{ if } 0 \le x < 10 \\ 4 + \cos(2\pi(x+2)^2), \text{ if } 10 \le x < 20 \\ 9 + \cos(2\pi x)\sin(2\pi x)^2, \text{ if } 20 \le x \le 30 \end{cases} \tag{40}$$

S is a non-stationary signal obtained by the concatenation of a toned (single frequency), a chirp, and a modulated signal. The signal S (40) and its piecewise constant approximations with $M = 2$ and $M = 3$ are represented in Fig. 2. Notice how different parts of the signal are approximated with its piecewise constant counterparts.

Thus, $\forall\ p \in [2;\ M]$ and $\forall\, k \in [0;\ N]$, $S_{M,N}$ is obtained by iterating the following scheme:

$$S_{p,k}[n] = \begin{cases} S_{p-1,l_p(k)}[n], \text{ if } 0 \le n \le l_p(k) \\ a_{l_p(k)+1,k}, \text{ if } l_p(k)+1 \le n \le k \end{cases} \tag{41}$$

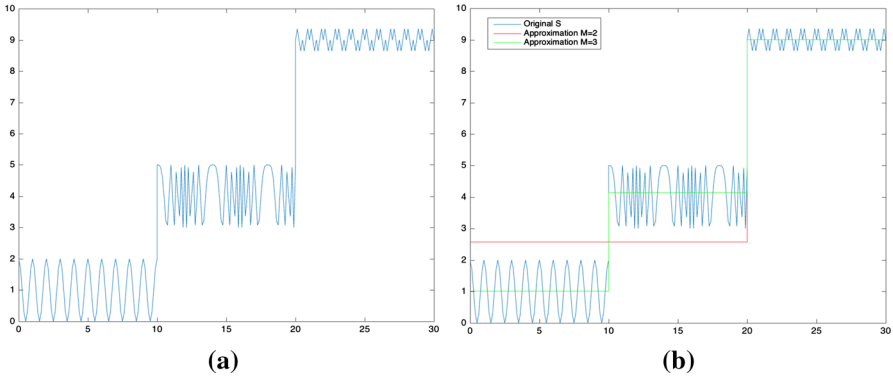


Fig. 2 **a** Signal S (40). **b** Piecewise constant approximation with our approach with $M = 2$ and $M = 3$

where $l_p(k) \in [0;\ k-1]$ is the value for which the sum $\varepsilon_{p-1,l} + c_{l+1,k}$ over all $l \in [0;\ k-1]$ is minimum and $a_{l_p(k)+1,k}$ are given by (23) in Proposition 4. Basically, Eq. (41) iteratively seeks for intervals on which S best fits to its mean in such intervals. The algorithm is detailed in Algorithm 1, and its complexity is O(MN^2) meaning that the proposed algorithm needs MN^2 operations in order to obtain the piecewise-based approximate S_{MN}. The proposed algorithm is looking for an optimal piecewise-based approximate signal S_{MN}, while quite simple, it is also efficient. M is a user input parameter that determines the number of piecewise constant approximation functions on M subdomains $(D_i)_{i=1}^{M}$. In every subdomain D_i, the corresponding piecewise constant function approximating S on D_i is calculated by taking means in an iterative way thanks to (41). Thus, the algorithm converges. As for any approximation method, M should be smaller than N. But, the closer the M to N, the higher the accuracy (*i.e.*, the lower the approximation error). For example, for an N samples piecewise constant signal S, one gets a perfect reconstruction (*i.e.*, approximation error equals to zero) for $M \approx N$.

4.2 Numerical Results

To show the efficiency of our proposed approach, we test our piecewise constant approximation algorithm on various signals and compare obtained results both qualitatively and quantitatively with Haar-based wavelet decompositions. For the wavelet-based decomposition, both linear approximations (15) and nonlinear approximations (19) are performed. The quantitative comparisons are performed by computing approximation errors using the L^2 norm and the minimum squared error (MSE).

The first tested signal has 128 samples and is given as

$$S_1 : [0;\ 1] \mapsto \mathbb{R};\ x \longrightarrow S_1(x) = 4\sin(20\pi x)\sin\left(\frac{\pi x}{5}\right) + 3\sin(10\pi x). \tag{42}$$

The signal S_1 (42) is plotted in Fig. 3a, and approximation results using our approach are displayed in Fig. 3a, with $M \in \{20, 40, 90\}$. Linear and nonlinear Haar-based

Algorithm 1: Proposed piecewise constant approximation algorithm

Input : Signal S, M and N.
Output: Approximation $S_{M,N}$.

```
foreach k = 1 to N do
    foreach l = 1 to k do
        Compute a_{k,l} (23) and c_{l,k} (22)
    end
end
foreach p = 1 to M do
    Compute S_{p,0} (34)
end
foreach k = 0 to N do
    Compute S_{1,k} (35)
end
foreach p = 2 to M do
    foreach k = 1 to N do
        Find l* = arg min_{l∈[0; k−1]} [ε_{p−1,l} + c_{l+1,k}]
    end
    foreach k = 2 to N do
        Compute S_{p,k} (41)
    end
end
```

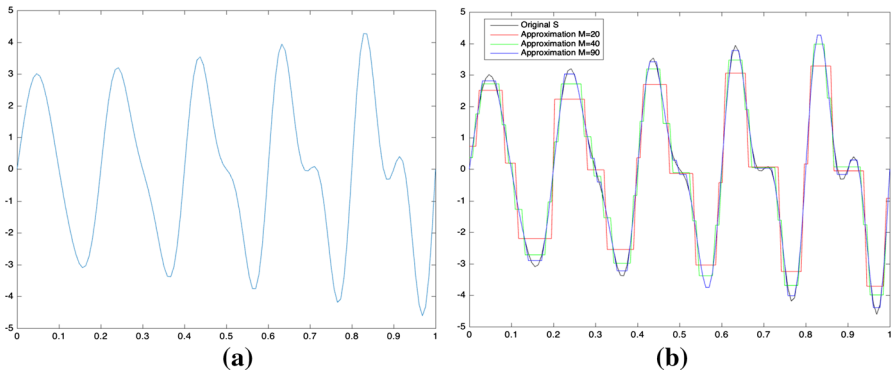
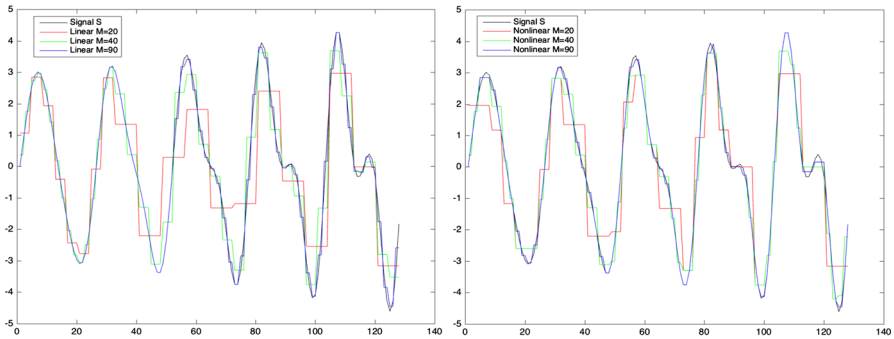


Fig. 3 **a** Signal S_1 (42). **b** Our approximation approach with $M = 20, 40$, and 90

wavelet approximations are, respectively, shown in Fig. 4a, b, with $K \in \{20, 40, 90\}$ also.

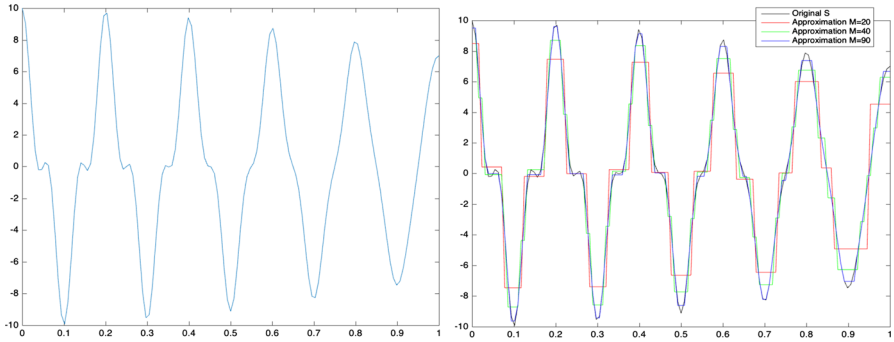
We also tested the approach on another signal with 128 samples, which is a modified version of S_1 in terms of amplitude and frequency of the modulated component. It is given as

$$S_2[0;\ 1] \mapsto \mathbb{R};\ x \longrightarrow S_2(x) = 3\cos(30\pi x)\cos\left(\frac{\pi x}{2}\right) + 7\cos(10\pi x). \tag{43}$$



(a) **(b)**

Fig. 4 Haar-based wavelet approximations with $K = 20, 40$, and 90. **a** Linear approximations (15). **b** Nonlinear approximations (19)



(a) **(b)**

Fig. 5 **a** Signal S_2 (43). **b** Our approximation approach with $M = 20, 40$ and 90

S_2 (43) is displayed in Fig. 5a, and approximation results using our approach are shown in Fig. 5a, with $M \in \{20, 40, 90\}$. Linear and nonlinear Haar-based wavelet approximations are, respectively, displayed in Fig. 6a, b, with $K \in \{20, 40, 90\}$ also.

The third tested signal is non-stationary signal of 128 samples, and is given as:

$$S_3 : [0;\ 1] \mapsto \mathbb{R};$$
$$x \longrightarrow S_3(x) = \left(\frac{x}{2} + 7\right) \sin(2\pi\sqrt{30x + 8}) + (5\sqrt{x} - 4) \sin\left(6\pi\left(3 + \frac{x}{2}\right)^2\right). \tag{44}$$

The S_3 signal (44) is plotted in Fig. 7a, approximation results using our approach are displayed in Fig. 7a, with $M \in \{20, 40, 90\}$. Linear and nonlinear Haar-based wavelet approximations are, respectively, shown in Fig. 8a, b, with $K \in \{20, 40, 90\}$ also.

The next tested signal is composed of two tones and has 256 samples; it is given as

$$S_4 : [0;\ 2] \mapsto \mathbb{R};\ x \longrightarrow S_4(x) = 2\sin(20\pi x) + 3\sin(2\pi x). \tag{45}$$

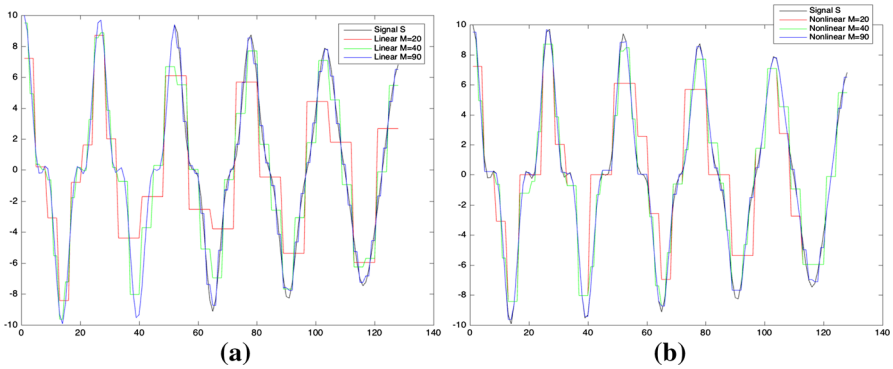


Fig. 6 Haar-based wavelet approximations with $K = 20, 40$ and 90. **a** Linear approximations (15). **b** Nonlinear approximations (19)

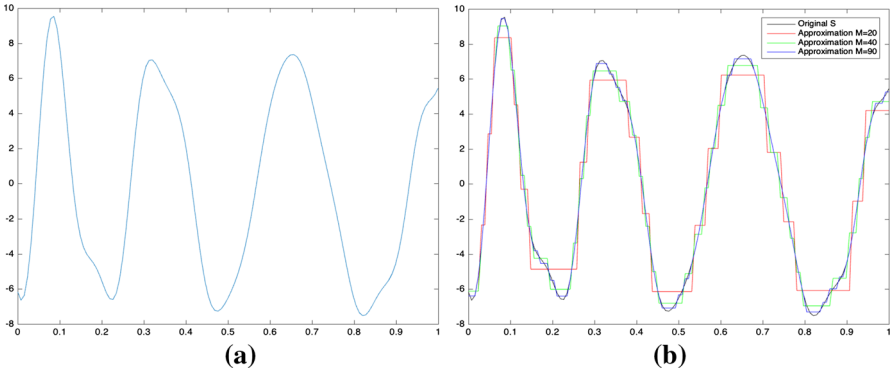


Fig. 7 **a** Signal S_3 (44). **b** Our approximation approach with $M = 20, 40$, and 90

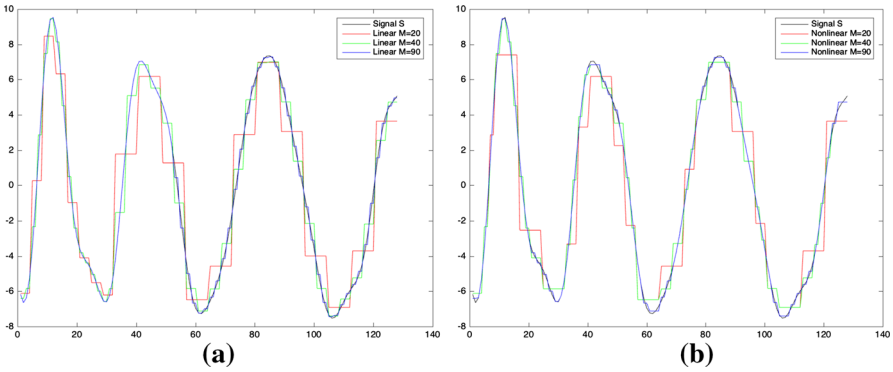


Fig. 8 Haar-based wavelet approximations with $K = 20, 40$, and 90. **a** Linear approximations (15). **b** Nonlinear approximations (19)

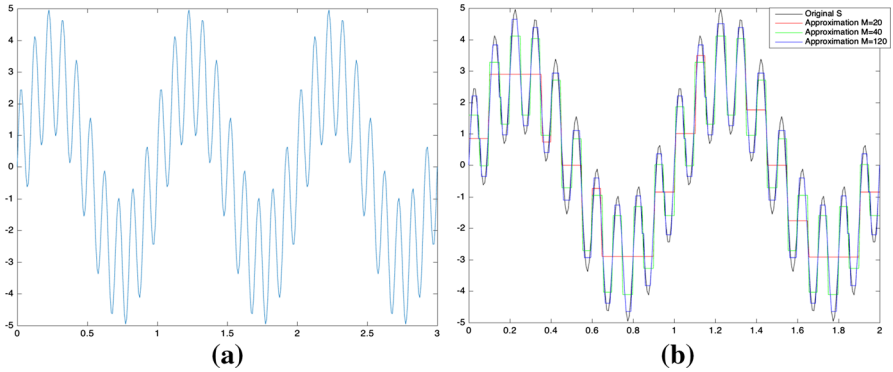


Fig. 9 **a** Signal S_4 (45). **b** Our approximation approach with $M = 20, 40$, and 120

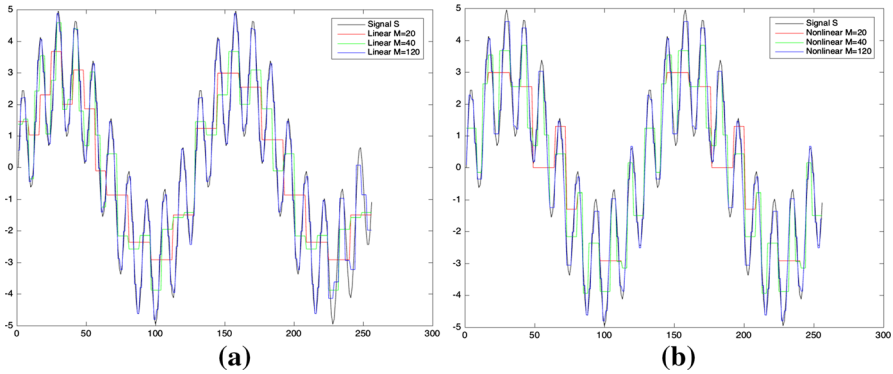


Fig. 10 Haar-based wavelet approximations with $K = 20, 40$, and 120. **a** Linear approximations (15). **b** Nonlinear approximations (19)

S_4 (45) is represented in Fig. 9a, and approximation results using our approach are shown in Fig. 9a, with $M \in \{20, 40, 120\}$. Linear and nonlinear Haar-based wavelet approximations are, respectively, displayed in Fig. 10a, b, for $M \in \{20, 40, 120\}$ also.

We lastly apply our approach to the 128 samples piecewise polynomial signal S_5 represented in Fig. 11a, and approximation results using our approach are shown in Fig. 11a, with $M \in \{20, 40, 120\}$. Linear and nonlinear Haar-based wavelet approximations are, respectively, displayed in Fig. 12a, b, with $K \in \{20, 40, 120\}$ also.

The above qualitative results show the efficiency of the proposed approximation method. In addition, we report in Table 1 the quantitative results with L^2 norm and the MSE metrics. The obtained values confirm that we obtain better approximations. Approximation errors of tested signals using our approach are always lower than the Haar-based wavelet linear and nonlinear approximations.

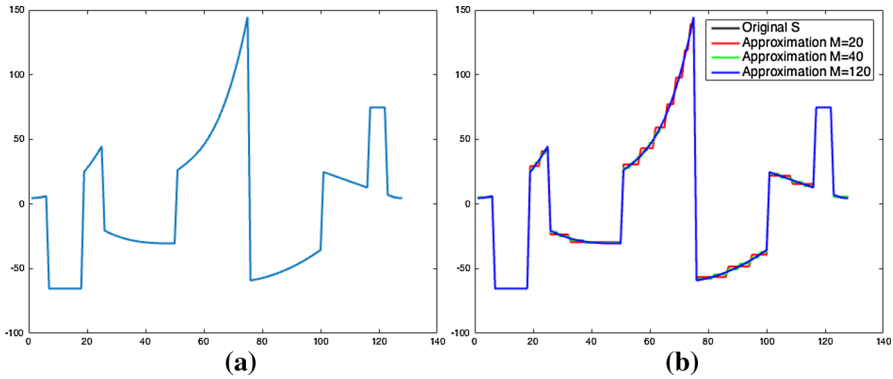


Fig. 11 **a** Representation of the signal S_5. **b** Proposed approximation approach with $M = 20, 40$, and 120

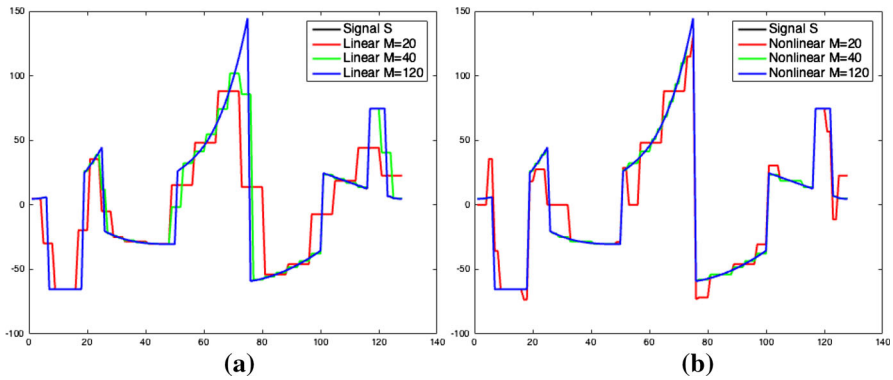


Fig. 12 Haar-based wavelet approximations with $K = 20, 40$, and 120. **a** Linear approximations (15). **b** Nonlinear approximations (19)

5 Conclusion

In this work, we proposed a nonlinear and optimal approach to approximate signals with piecewise constant functions. This is achieved by minimizing over the signal subdomains the minimum squared error between the signal and its corresponding piecewise approximations; the subdomains being determined by the number of piecewise approximation functions. The proposed algorithm is efficient thanks to an established and proven optimal recursive relationship. Our experimental results on various signals showed better approximations in comparison with Haar wavelet-based linear and nonlinear approximation methods. The complexity of the proposed approach is in the order of $O(MN^2)$, whereas the wavelet-based method is $(O(N \log N)$, with the number of parameters is correlated to the computational effort. It might be interesting to find a way to optimize the number M of piecewise functions (*i.e.*, the number of subdomains); however, adding this extra step in the proposed algorithm will increase the overall complexity of the method, and the computational memory will be an issue especially for big N values. As a future work, we need to consider robust approxima-

Table 1 Approximation errors with our proposed approach (1), linear (15), and nonlinear wavelet-based approximations (19)

		L^2 norm	MSE
Signal S_1 (42): $K = 40$	Proposed	4.3578	0.14721
	Linear	7.9839	0.49799
	Nonlinear	5.294	0.21896
$K = 90$	Proposed	1.0542	0.0086154
	Linear	3.3838	0.089455
	Nonlinear	1.4253	0.015871
Signal S_2 (43): $K = 40$	Proposed	10.5908	0.86949
	Linear	16.9341	2.2403
	Nonlinear	13.0215	1.3247
$K = 90$	Proposed	2.5344	0.049791
	Linear	6.2751	0.30763
	Nonlinear	3.636	0.10328
Signal S_3 (44): $K = 40$	Proposed	6.4353	0.32103
	Linear	9.9447	0.77264
	Nonlinear	7.6339	0.45529
$K = 90$	Proposed	1.5012	0.01747
	Linear	3.381	0.089305
	Nonlinear	1.7741	0.02459
Signal S_4 (45): $K = 40$	Proposed	10.3442	0.41635
	Linear	18.4797	1.334
	Nonlinear	14.1364	0.78062
$K = 120$	Proposed	4.133	0.066464
	Linear	6.6729	0.17394
	Nonlinear	5.129	0.10276
Signal S_5: $K = 20$	Proposed	29.0851	6.6089
	Linear	341.455	910.8713
	Nonlinear	146.5394	167.764
$K = 40$	Proposed	10.164	0.80709
	Linear	197.2544	303.9788
	Nonlinear	16.9871	2.2544
$K = 120$	Proposed	0	0
	Linear	1.039	0.0084335
	Nonlinear	3.0334e-13	7.1885e-28

tion strategy to handle noisy signals and extend it for two-dimensional digital images and to higher dimensional spaces. A denoising approach such as the one considered in [42] for piecewise constant signals could be used for the noise robustness purpose. We further plan to investigate the extension of the denoising combined with our approach to piecewise affine and polynomial approximations, which could yield better approximations than piecewise constant ones.

References

1. L. Ambrosio, N. Fusco, D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems (Oxford Mathematical Monographs)* (Oxford University Press, Oxford, 2000)
2. I.E. Auger, C.E. Lawrence, Algorithms for the optimal identification of segment neighborhoods. Bull. Math. Biol. **51**(1), 39–54 (1989). https://doi.org/10.1007/bf02458835
3. R. Bellman, R. Roth, Curve fitting by segmented straight lines. J. Am. Stat. Assoc. **64**(327), 1079–1084 (1969). https://doi.org/10.1080/01621459.1969.10501038
4. T. Blu, M. Unser, Quantitative Fourier analysis of approximation techniques: part I—interpolators and projectors. IEEE Trans. Signal Process. **47**(10), 2783–2795 (1999)
5. A. Bouchikhi, A. Boudraa, Multicomponent AM-FM signals analysis based on EMD-B-splines ESA. Signal Process. **92**(9), 2214–2228 (2012)
6. H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations* (Springer, Berlin, 2011)
7. J.D. Bruce, *Optimum quantization*. Technical report, Research Laboratory of Electronics. MIT (1965)
8. A.P. Calderón, Intermediate spaces and interpolation the complex method. Stud. Math. **24**, 113–190 (1964)
9. F. Chaplais, Generalization of the interaction between Haar approximation and polynomial operators to higher order methods. Anal. Theory Appl. Anal. **22**(4), 301–318 (2006)
10. D. Chen, S.B. Ko, A novel decimal logarithmic converter based on first-order polynomial approximation. Circuits Syst. Signal Process. **31**(3), 1179–1190 (2012)
11. R.J. Cintra, An integer approximation method for discrete sinusoidal transforms. Circuits Syst. Signal Process. **30**(6), 1481 (2011)
12. G. Cybenko, Approximation by superpositions of a sigmoidal functions. Math. Control Signals Syst. **2**, 303–314 (1989)
13. C. de Boor, *A Practical Guide to Splines* (Springer, Berlin, 1978)
14. R.A. DeVore, Nonlinear approximation. Acta Numerica **7**, 51–150 (1998)
15. R.A. DeVore, *Multiscale, Nonlinear and Adaptive Approximation* (Springer, Berlin, 2009), pp. 169–201. chap. 6
16. R.A. DeVore, G.G. Lorentz, *Constructive Approximation* (Springer, Heidelberg, 1993), pp. 354–389. chap. 12
17. S. Dhabu, A.P. Vinod, A new time-domain approach for the design of variable FIR filters using the spectral parameter approximation technique. Circuits Syst. Signal Process. **36**(5), 2154–2165 (2017)
18. N. Dobigeon, J.Y. Tourneret, Joint segmentation of piecewise constant autoregressive processes by using a hierarchical model and a Bayesian sampling approach. IEEE Trans. Signal Process. **55**(4), 1251–1262 (2007)
19. D.L. Donoho, Compressed sensing. IEEE Trans. Inf. Theory **52**(4), 1289–1306 (2006)
20. M. Elad, M. Aharon, Image denoising via sparse and redundant representations over learned dictionaries. Trans. Image Process. **15**(12), 3736–3745 (2006)
21. L.C. Evans, R.F. Gariepy, *Measure Theory and Fine Properties of Functions. Studies in Advanced Mathematics* (CRC Press, Boca Raton, 1992)
22. Z. Fan, L. Guan, Approximate $l0$-penalized estimation of piecewise-constant signals on graphs, pp. 1–34. ArXiv arXiv:1703.01421v2 (2017)
23. J. Frecon, N. Pustelnik, P. Abry, L. Condat, On-the-fly approximation of multivariate total variation minimization. IEEE Trans. Signal Process. **64**(9), 2355–2364 (2016)
24. J.H. Friedman, Greedy function approximation: a gradient boosting machine. Ann. Stat. **29**(5), 1189–1232 (2001)
25. A.J. Gibberd, J.D.B. Nelson, Regularized estimation of piecewise constant Gaussian graphical models: The group-fused graphical Lasso. J. Comput. Graph. Stat. **26**(3), 623–634 (2017)
26. E. Giusti, *Minimal Surfaces and Functions of Bounded Variation* (Birkhuser, Basel, 1994)
27. A. Grossmann, J. Morlet, Decomposition of Hardy functions into square integrable wavelets of constant shape. SIAM J. Math. Anal. **15**(4), 723–736 (1984)
28. E.M. Hamed, A.M. AbdelAty, L.A. Said, A.G. Radwan, Effect of different approximation techniques on fractional-order KHN filter design. Circuits Syst. Signal Process. **37**, 1–31 (2018)
29. H.P. Hiriyannaiah, G.L. Bilbro, W.E. Snyder, R.C. Mann, Restoration of piecewise-constant images by mean-field annealing. J. Opt. Soci. Am. A **6**(12), 1901–1912 (1989)

30. N.E. Huang, Z. Shen, S.R. Long, M.C. Wu, H.H. Shih, Q. Zheng, N.C. Yen, C.C. Tung, H.H. Liu, The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis. R. Soc. **454**, 903–995 (1998)
31. Y. Isaac, Q. Barthélemy, C. Gouy-Pailler, M. Sebag, J. Atif, Multi-dimensional signal approximation with sparse structured priors using split Bregman iterations. Signal Process. **130**, 389–402 (2017)
32. B. Jackson, J. Scargle, D. Barnes, S. Arabhi, A. Alt, P. Gioumousis, E. Gwin, P. San, L. Tan, T.T. Tsai, An algorithm for optimal partitioning of data on an interval. IEEE Signal Process. Lett. **12**(2), 105–108 (2005). https://doi.org/10.1109/lsp.2001.838216
33. A.C. Jensen, A.S. Solberg, Fast hyperspectral feature reduction using piecewise constant function approximations. IEEE Geosci. Remote Sens. Lett. **4**(4), 547–551 (2007)
34. J.P. Kahane, *Teoria constructiva de functiones. Course notes* (University of Buenos Aires, Viamonte, 1961)
35. J. Kang, H. Jung, H.N. Lee, K. Kim, One-dimensional piecewise-constant signal recovery via spike-and-slab approximate message-passing, in *48th Asilomar Conference on Signals, Systems and Computers* (Pacific Grove, CA, USA, 2014), pp. 1458–1462
36. H. Konno, T. Kuno, Best piecewise constant approximation of a function of single variable. Oper. Res. Lett. **7**(4), 205–210 (1988). https://doi.org/10.1016/0167-6377(88)90030-2
37. S.C. Kramer, H.W. Sorenson, Recursive Bayesian estimation using piece-wise constant approximations. Automatica **24**(6), 789–801 (1988)
38. B.I. Kvasov, *Methods of Shape Preserving Spline Approximations* (World Scientific, Singapore, 2000)
39. F. Leulmi, Y. Ferdi, Improved digital rational approximation of the operator s^{α} using second-order s-to-z transform and signal modeling. Circuits Syst. Signal Process. **34**(6), 1869–1891 (2015)
40. Q. Li, B. Shen, Z. Wang, F.E. Alsaadi, A sampled-data approach to distributed h ∞ resilient state estimation for a class of nonlinear time-delay systems over sensor networks. J. Franklin Inst. **354**(15), 7139–7157 (2017). https://doi.org/10.1016/j.jfranklin.2017.08.036
41. Q. Li, B. Shen, Z. Wang, F.E. Alsaadi, An event-triggered approach to distributed h∞ state estimation for state-saturated systems with randomly occurring mixed delays. J. Franklin Inst. **355**(6), 3104–3121 (2018). https://doi.org/10.1016/j.jfranklin.2018.02.007
42. M.A. Little, N.S. Jones, Generalized methods and solvers for noise removal from piecewise constant signals. II. New methods. Proc. R. Soc. A Math. Phys. Eng. Sci. **467**(2135), 3115–3140 (2011). https://doi.org/10.1098/rspa.2010.0674
43. G. Makkena, M. Srinivas, Nonlinear sequence transformation-based continuous-time wavelet filter approximation. Circuits Syst. Signal Process. **37**(3), 965–983 (2018)
44. S. Mallat, *A Wavelet Tour of Signal Processing* (Academic Press, Cambridge, 1999)
45. S. Mallat, *A Wavelet Tour of Signal Processing: The Sparse Way* (Elsevier, Amsterdam, 2009)
46. S. Mallat, Z. Zhang, Matching pursuits with time-frequency dictionaries. IEEE Trans. Signal Process. **41**(12), 3397–3415 (1993)
47. R. Martin, W. Shen, Asymptotically optimal empirical Bayes inference in a piecewise constant sequence model. Technical Report arXiv:1712.03848v1 (2017)
48. H. Min, W. Jia, X.F. Wang, Y. Zhao, Y.T. Luo, A polynomial piecewise constant approximation method based on dual constraint relaxation for segmenting images with intensity inhomogeneity. Pattern Recognit. **73**, 15–32 (2018)
49. D.B. Mumford, J. Shah, Optimal approximations by piecewise smooth functions and associated variational problems. Commun. Pure Appl. Math. **42**(5), 577–685 (1989)
50. M. Nikolova, M.K. Ng, S. Zhang, W.K. Ching, Efficient reconstruction of piecewise constant images using nonsmooth nonconvex minimization. SIAM J. Imaging Sci. **1**(1), 2–25 (2008)
51. G. Ongie, S. Biswas, M. Jacob, Convex recovery of continuous domain piecewise constant images from nonuniform Fourier samples. IEEE Trans. Signal Process. **66**(1), 236–250 (2018)
52. Y.C. Pati, R. Razaiifar, P.S. Krishnaprasad, Orthogonal matching pursuit: recursive function approximation with applications to wavelet decomposition, in *27th Annual Asilomar Conference on Signals Systems and Computers*, pp. 1–5 (1993)
53. M.J.D. Powell, *Approximation Theory and Methods* (Cambridge University Press, Cambridge, 1981)
54. S. Qian, D. Chen, K. Chen, Signal approximation via data-adaptive normalized Gaussian functions and its applications for speech processing, in *IEEE ICASSP*, vol. 1, pp. 141–144. San Francisco, CA, USA (1992)
55. S. Qian, D. Chen, Q. Yin, Adaptive chirplet based signal approximation, in *IEEE ICASSP*, pp. 1781–1784. Seattle, US (1998)

56. S.D. Roy, A new Chebyshev-like low-pass filter approximation. Circuits Syst. Signal Process. **29**(4), 629–636 (2010)
57. R. Rubinstein, M. Zibulevsky, M. Elad, Double sparsity: Learning sparse dictionaries for sparse signal approximation. Trans. Signal Process. **58**(3), 1553–1564 (2010)
58. J.A. Sanders, F. Verhuls, *Averaging Methods in Nonlinear Dynamical Systems* (Springer, Berlin, 1985)
59. I. Selesnick, M. Farshchian, Sparse signal approximation via non-separable regularization. IEEE Trans. Signal Process. **65**(10), 1–15 (2017)
60. J.G. Serra, M. Testa, R. Molina, A.K. Katsaggelos, Bayesian K-SVD using fast variational inference. IEEE Trans. Image Process. **26**(7), 3344–3359 (2017)
61. J. Shah, I. Qureshi, Y. Deng, K. Kadir, Reconstruction of sparse signals and compressively sampled images based on smooth $l1$-norm approximation. J. Signal Process. Syst. **88**, 333–344 (2017)
62. R. Tibshirani, Regression shrinkage and selection via the Lasso. J. R. Stat. Soc. Ser. B (Methodol.) **58**(1), 267–288 (1996)
63. M. Unser, A. Aldroubi, M. Eden, B-spline signal processing: part I—theory. Trans. Signal Process. **41**(2), 821–832 (1993)
64. M. Unser, A. Aldroubi, M. Eden, B-spline signal processing: part II—efficient design and applications. Trans. Signal Process. **41**(2), 834–848 (1993)
65. M. Unser, P.D. Tafti, Stochastic models for sparse and piecewise-smooth signals. IEEE Trans. Signal Process. **59**(3), 989–1006 (2011)
66. F. van Belzen, S. Weiland, Reconstruction and approximation of multidimensional signals described by proper orthogonal decompositions. IEEE Trans. Signal Process. **56**(2), 576–587 (2008)
67. S. Vovk, V. Borulko, Determination of amplitude levels of the piecewise constant signal by using polynomial approximation. Radioelectron. Commun. Syst. **60**(3), 141–153 (2017)
68. B. Weng, K.E. Barner, Optimal signal reconstruction using the empirical mode decomposition. EURASIP J. Adv. Signal Process. **2008**, 1–12 (2008)
69. S.J. Wright, R.D. Novak, M.A.T. Figueiredo, Sparse reconstruction by separable approximation. IEEE Trans. Signal Process. **57**(7), 2479–2493 (2009)
70. X. Zhou, J. Liu, X. Wan, W. Yu, Piecewise-constant and low-rank approximation for identification of recurrent copy number variations. Bioinformatics **30**(14), 1943–1949 (2014)

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